A note on Open AI's "A counterexample to Hadwiger's conjecture"

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1 Summary

In the note they construct a triangle-free graph G that has many 4-cycles, such that for the complement graph we has its independence number $\alpha(\bar{G}) = 2$ and connected matching number $\mathrm{cm}(\bar{G}) < m/100$ where m is number of vertices. The connected matching number is defined as follows:

Definition 1. *A matching M in a graph H is called a* connected matching *if every two distinct edges of M are joined by an edge of H: for all distinct $e, f \in M$, there exist $u \in V(e)$ and $v \in V(f)$ such that $uv \in E(H)$. The* connected matching number *is*

$$\mathrm{cm}(H) = \max\{|M| : M \text{ is a connected matching in } H\}.$$

The complement graph provides a counterexample to Hadwiger's conjecture, which asserts that $\chi(H) \le h(H)$, where $h(H)$ is the largest integer t such that H contains a K_t minor. Indeed, Proposition 3.5 of the paper (proved in Appendix A) gives

$$h(H) \le \frac{|V(H)| + 4\,\mathrm{cm}(H) + 2}{3}.$$

Applying this to $H = \bar{G}$, and using $\alpha(\bar{G}) \le 2$ so that every color class has at most two vertices, we obtain, for m large enough,

$$h(\bar{G}) \le \frac{m + 4\,\mathrm{cm}(\bar{G}) + 2}{3} < \frac{26m}{75} + \frac{2}{3} < \frac{m}{2} \le \chi(\bar{G}).$$

Thus $h(\bar{G}) < \chi(\bar{G})$, contrary to Hadwiger's conjecture.

A The clique-minor bound (Proposition 3.5)

Proposition 1. *For every finite nonempty graph H with m vertices,*

$$h(H) \le \frac{m + 4\,\mathrm{cm}(H) + 2}{3}.$$

Proof. Write $c = \mathrm{cm}(H)$, $b = h(H)$. Thus that there are b pairwise disjoint, nonempty connected vertex sets, with an edge joining every pair of sets. These sets are the branch sets of a K_b minor. Let s be the number of sets with one vertex and e be the number of sets with two vertices.

Firstly $c > e + \frac{s-1}{2}$: Each two-vertex branch set has an edge. And pairs of singleton branches supports an edge, the set of them is a connected matching, so we have the desired inequality.

Then every remaining branch set has at least three vertices. Since the branch sets are disjoint,

$$m \geq s + 2e + 3(b - s - e) = 3b - 2s - e.$$

Combining two inequality give use the contradiction. □